

► Introduction

In August 2023, the ILO published the first global employment estimates of potential occupational exposure to generative artificial intelligence (GenAI) (Gmyrek et al., 2023). This research was prompted by concerns – that have featured prominently in the media – of the potential for the technology to replace large shares of knowledge work.

Since the release of the 2023 estimates, there have been further technological advances in GenAI, including the introduction of Large Language Models (LLMs) with expanded capabilities in image recognition and audio and video production. Moreover, many new GenAI applications have been developed as a result of the possibility of linking GenAI models with Application Programming Interface (API) tools. Indeed, GenAI has been likened to a general-purpose technology, meaning that it may continue generating new applications and innovations across various sectors, driving long-term economic and societal transformations.

Despite this rapidly developing landscape, it is important for policy makers to develop deeper understanding of how technological advances might impact existing jobs. To address this need, the Research department of the ILO has partnered with NASK, the National Research Institute of the Ministry of Digital Affairs in Poland, to develop a more precise assessment of potential effects of GenAI on employment across countries.

This study builds on the method developed by the ILO (Gmyrek, Berg and Bescond, 2023 - GBB hereafter), which used task descriptions from the 4-digit International Standard Classification of Occupations (ISCO-08) and the GPT-4 model to estimate task-level and occupational scores of exposure to GenAI technology and then derives global employment estimates based on official ILO labour force data. But the study includes several key refinements.

The first step in our analysis is based on Poland's national 6-digit classification of occupations that includes nearly 30,000 tasks. This expands tenfold the number of tasks given in the ISCO-08 structure and enables a fine-grained assessment of tasks' automation potential, based on a more up-to-date set of occupations and tasks than those captured in the ISCO-08 system. As Poland's 6-digit classification is aligned with the 4-digit ISCO-08, the occupational scores can be compiled at the 4-digit level allowing a straightforward comparison and updating of the 2023 estimates of the ILO (GBB). In addition, the study improves the methodological framework of GBB, by drawing on the combined strengths of human and AI abilities, with an additional layer of expert verification.

Specifically, we surveyed 1,640 people currently employed in Poland in each ISCO-08 1-digit group to rank the automation potential of a representative sample of tasks that pertain to their occupational group. As the second step, we asked a mix of national and international experts to conduct a detailed review of a large sample of tasks previously assessed in the survey to validate or adjust the level of automation scores through an iterative process. Since we maintain the hierarchical link to ISCO-08 and focus on technological feasibility rather than specific country context, by combining opinions of local and international experts, we create a system that enables more precise predictions of the potential of automation of individual occupational tasks with GenAI technology, including at the international level. To achieve that, we input the human scores into an AI model, to generate scores reflecting human judgement for nearly 30,000 individual tasks pertaining to 2,500 occupations at 6-digit level in Poland. We then use the same AI predictor to re-generate scores for the tasks in ISCO-08, previously provided by GBB (2023), thereby adjusting the 2023 index of the ILO, which now benefits from both the input of people

employed in each ISCO-08 1-digit group and from expert opinions. As the final step, we provide updated global, regional and income-based estimates of employment that has the potential to be transformed by Gen AI technology.

Our study specifically targets GenAI tools, isolating the effects of this recent family of AI from broader AI applications, such as the use of Machine Learning (ML) algorithms or image recognition in industrial production. There are several considerations that justify such selective focus. First, we respond to the growing need for a research tool that could provide a better picture of GenAI's potential impact on national labour markets, given the growing societal angst concerning this technology and the recent reports about increasing levels of adoption of GenAI at work (Bick et al., 2024; Faverio and Tyson, 2022; Grampp et al., 2023; Maison & Partners and ThinkTank, 2024; Pew Research Center, 2023; Rutgers, 2024). Second, while several indicators of occupational exposure to broader AI technologies are available to researchers (Nurski and Vansteenkiste, 2024), very few tools allow for a more detailed focus on GenAI (Nurski and Ruer, 2024). A new, more precise exposure index enables a closer alignment of the academic work on digital economy and labour with the significant interest in the GenAI technology in the media and in the public debate.¹ Third, the methodological blueprint provided by our research can be quite flexibly expanded to other types of AI in the future or focused more narrowly on specific subsets of digital technologies and sectors of their application.

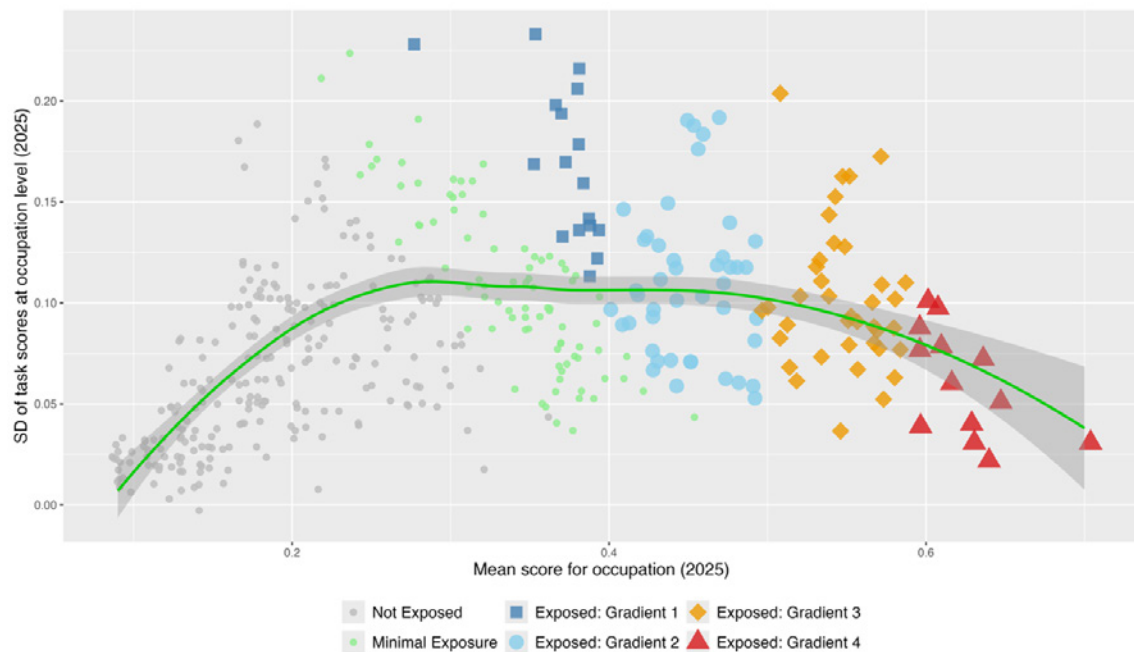
This paper is the first in a series of upcoming publications based on ILO and NASK cooperation that strive to improve the methodology for assessing employment effects of AI, but also to document in detail the methods used, including the survey, expert verification process and the construction of the AI model used for predictions. Subsequent papers will present an assessment of the potential impact of GenAI on the Polish labour market, based on more detailed occupation and task-level data and using a mix of quantitative and qualitative methods. All publications will include the methodological blueprints and technical details of both the quantitative and qualitative components, including survey questionnaires and interview guidelines. By making these tools openly accessible, we hope to contribute to the development of an improved method for assessing potential impact of GenAI on jobs and to stimulate more detailed national-level studies, including in lower- and middle-income countries.

¹ For example, Felten et al. (2023) provided an updated version of their original exposure index from 2021, adjusting for the abilities of GenAI tools. As shown in their paper, the adjustment does not result in a major shift in occupations exposure and still makes it impossible to assess the abilities of this new technology in a selective manner. See Gmyrek et al. (2024) for a more detailed comparison of these scores.

► 8 Adjusted Global Index of GenAI Exposure

To classify ISCO-08 occupations into varying levels of exposure to Generative AI (GenAI), we update the framework introduced in Gmyrek et al. (2023). Similarly to the original approach, we rely on two moments of distribution: occupational mean and standard deviation (SD) of task-level scores pertaining to a given 4-digit occupation. However, instead of using automation and augmentation potential as two extreme categories with the "big unknown" in between, we now adopt a more nuanced classification scheme that represents a spectrum of exposure, with occupations considered as exposed classified into four increasing gradients (Figure 16 and Table 5).

► Figure 16. AI Exposure Gradients: ISCO-08 4-digit level occupations



Note: Figure 16 presents all ISCO-08 4-digit occupations classified into GenAI exposure categories as per the criteria in Table 5. A detailed list of occupation-level means and SDs is available in the Appendix, Table A1. In addition, task-level scores within the overall hierarchy of ISCO-08 are available here: https://pgmyrek.github.io/2025_GenAI_scores_ISCO08/

Gradient 1 represents occupations with low overall GenAI exposure but significant variability across tasks (Low exposure, high task variability). While some tasks within these roles may have high automation potential, the occupation has many tasks that continue requiring human roles, aligning closely with the notion of augmentation in the old framework. Gradient 2 includes occupations with moderate GenAI exposure and a mix of highly exposed and minimally exposed tasks, resulting in uneven impacts where some tasks may be disrupted while others remain unaffected (Moderate exposure, high task variability). Gradient 3 captures occupations where a significant portion of tasks are consistently exposed to GenAI, signaling growing automation risks and requiring adaptation strategies for workers (Significant exposure, high task variability). Gradient 4 highlights occupations with the highest share of tasks exposed to potential GenAI-driven automation, and with a high consistency of this exposure across tasks (Highest exposure, low task variability). This gradient aligns closely with the notion of automation risk in our earlier classification system (Gmyrek et al., 2023). In addition, among occupations outside these four categories, we introduce a more granular distinction of Minimal Exposure, where some interaction with GenAI